

ICT
0417 Past Papers

Chapter 4
**Networks and the effects of
using them**

- 8 A company has an extranet.

Explain what is meant by an extranet.

Three from:

Extranet is a network controlled by a company

Restricted/private network

Allows access for partners/vendors/suppliers

Allows access for authorised customers

[3]

- 10 A headteacher is planning to hold a video-conference with headteachers from other schools.

The headteachers already have basic internet connected desktop computer systems including a monitor, keyboard and mouse.

- (a) Identify **three** pieces of additional hardware they need in order to participate in a video-conference.

1

2

3

10(a)

Microphone

Video camera/webcam

Speakers/headphones

[3]

- (b) A technician is setting up the hardware for the video-conference.

State **three** tasks the technician would need to do to set up the hardware for the video-conference.

You can assume that the hardware required has already been purchased.

1

2

3

10(b)

Three from:

Ensure the video camera/webcam, microphone, speakers/headphones are switched on

Ensure the hardware devices are connected

Carry out tests on the microphone, video camera/webcam and speakers/headphones

Adjust the video camera/webcam so the headteacher can be seen

Check the volume levels

Check internet connectivity

[3]

14 Compare the similarities and differences between Bluetooth and wi-fi.

14

Max **three** from:

Similarities

Both use wireless communications

Both use radio frequencies for communication

Both allow several devices to be connected

Both use security when sending data

Max **three** from:

Differences

wi-fi has a faster data transfer rate

wi-fi has a greater range of transmission

wi-fi uses key matching encryption

Bluetooth uses WEP and WPA security together

wi-fi can be broadcast

wi-fi connects more devices

Bluetooth has a lower frequency than wi-fi

[4]

(d) The data stored on the computer system needs to be protected from unauthorised access.

Discuss the effectiveness of different methods of increasing security of this data.

7(d)

Eight from:

Factors increasing effectiveness

Strong passwords are difficult to crack
 Biometric passwords are harder to crack
 Regularly changing passwords increases security
 Use of two-factor authentication
 Using different passwords for parts of the computer system makes it more difficult to gain access to the full system
 Firewall required to stop attacks from computers
 Firewall stops fraudulent sites attacking the computer
 Anti-spyware stops passwords being seen when typed in

Factors reducing effectiveness

Too complex a password can be easily forgotten
 Passworded files may not be backed up
 Using a number of different passwords can become cumbersome
 Regularly changing passwords means that passwords may be forgotten
 May be difficult to choose unique password if it is changed every few weeks
 Passwords may become more easy to guess if regularly changed
 Hackers can breach most passwords
 The firewall can affect the operation of the computer and stop some uploads
 The firewall may need to be shut down at times to upload files therefore making computer unsafe
 Some legitimate software can be blocked by the firewall

To gain full marks both sides of the discussion are needed

Act
Go to

Activate Win
[8]
Go to Settings t

- 8 Compare and contrast the features of a switch and a hub.

8

Six from:**Comparison**

Both are network devices

Both are hardware devices

Both have computers and devices connected to them

Both send data to devices//Both are communication devices

Contrast

In a hub data packets/data are broadcast to every device connected to it

A hub is less secure than a switch in distributing data

Switch checks the data packet

Switch sends to an appropriate device

To gain full marks both contrast and comparison is needed

[6]

- 4 Complete the sentences below using the most appropriate word from the list.

cookies

hacking

pharming

phishing

smishing

spam

spyware

virus

- (a) The act of gaining unauthorised access to a computer system is called

[1]

- (b) The software that gathers data by monitoring key presses on a user's keyboard is called

[1]

- (c) A fake text message, that could contain a link, sent to a mobile phone is called

[1]

4(a)	Hacking
4(b)	Spyware
4(c)	Smishing

9 (a) Describe two uses of a company intranet.

1.....	9(a)	Two from: Send/receiving emails Send/receive attachments/files Internal file transfer within the company Allows the creation of extranets Surf internal information/webpages Sharing/use of applications/resources	
.....			
2.....			
.....			

[2]

(b) Social networking is used to communicate with friends; this is one use of the internet.

Describe four other uses of the internet.

1.....	9(b)	Four from, with descriptions: Email Use of the web Newsgroups Online chat Streaming Internet telephone Online conferencing Gaming File transfer	
.....			
2.....			
.....			
3.....			
.....			
4.....			
.....			

[4]

- (c) A major part of the everyday use of the internet is social networking. Currently some social networks are not policed. The owners of a website have full discretion on what material is removed and what is allowed to be kept.

Discuss the benefits and drawbacks of policing social networking sites.

Benefits

Max four:

- Can reduce cyber bullying/hate speech/sexual harassment
- Can reduce social media from being used for illegal activities
- Can reduce the ability to incite violence/trolling
- Could prevent people from being subjected to inappropriate messages
- Can reduce illegal/inappropriate images

Drawbacks

- Freedom of speech is reduced
- Would be very difficult to police as there is a massive amount of information
- Different rules in different countries and social media crosses borders
- Very costly to create an effective deterrent
- Could force individuals to use other un-policed methods of communication
- Who decides what is acceptable and what is not acceptable
- Privacy issues as people checking could read messages
- Difficult to police as users can be anonymous

To gain full marks there needs to be benefits and drawbacks

[8]

- 7 A family has purchased a wireless router for use in their home to allow their devices to be connected via WiFi to form a wireless local area network (WLAN).

- (a) When the family purchased the wireless router they were advised to change the default password.

Explain why they should change the default password.

- 7(a) **Two from:**
 So that other people cannot access the router using the known default password
 As it is wireless the password could be cracked from outside easily and then used for free by others
 So that people cannot make changes to/access personal data on the router
 It would increase security of the data

[2]

- (b) Describe the process of using WiFi to connect to a WLAN.

- Four from:**
 The computer/device sends data to its wireless adapter
 The data is converted into radio waves
 The radio waves are transmitted
 A wireless adapter receives the radio waves
 Radio waves are converted into data
 The computer initiates a handshake with the wireless router
 The wireless router responds
 Establishes a session/pairing
 Wireless router checks the password/key is correct
 Data is transmitted
 User has to enter a password/passkey/automatic entry of key for known connections

[4]

- (c) Their son produces large complex diagrams on his desktop computer at home. He needs to be able to access the diagrams when he is at university. Even if reduced in file size, the diagrams would be too large to send using email.

Describe in detail how he could transfer the diagrams from his home desktop computer to be able to access them at university, without using additional hardware.

7(c)

Four from:

Create a cloud storage account
 Ensure sufficient storage in the cloud account
 Log into the cloud at home
 Upload/transfer the diagrams from his home computer to the cloud
 At university: access/login to his cloud storage account
 At university: download/access the stored diagrams

[4]

- (d) One of the parents in the family works for a government department and sends sensitive data from their home computer to the office. The data is encrypted before being sent to ensure it is harder to understand if it is intercepted.

Describe **four** other preventative measures that could be taken to avoid network security issues.

1.....

2..... 7(d)

Four from:

Install security devices/firewalls
 Use a VPN
 Install security software/anti-virus/anti-spyware
 Set up security settings in the router/operating system
 Data backup in case of data loss/corruption
 Restrict access to the network infrastructure
 Train the family in the safe and secure use of the equipment
 Not opening emails/attachments from unknowns/phishing emails
 Not downloading files from unknowns/phishing emails
 Use a dongle to restrict use of software

3.....

4.....

[4]

- 7 Bluetooth is a method of communication to connect devices together.

Describe how a smartphone can use Bluetooth to connect to another device.

7

Four from:

Smartphone sends data to its Bluetooth connector
 Data is converted to radio waves
 The smartphone automatically detects nearby devices
 The smartphone pairs to the other device if in range
 The smartphone randomly chooses one of 79 possible channels
 If the channel is in use, it randomly chooses another
 The connection changes channels, every few seconds
 Sends the radio wave to the other device

[4]

- 8 An organisation uses the internet and an intranet.

(a) Explain how the purpose of the internet differs from the purpose of an intranet.

8(a)

Four from:

Internet is used to share and distribute information outside the organisation
 whereas the intranet within an organisation
 Internet is used to send and receive emails and messages outside the
 organisation whereas intranet sends and receives messages within an
 organisation
 The intranet meets the needs of the organisation whereas internet aims to
 meet the needs of everybody
 The internet facilitates collaboration with others whereas the intranet is only
 used to collaborate with other members within an organisation
 The internet is used to browse the World Wide Web

[4]

Employees in the organisation are allowed to access the intranet using their smartphones.

The employees are required to secure their smartphones using either facial recognition, passwords or voice recognition.

- (b) Tick (✓) whether the following statements refer to **facial recognition**, **password** or **voice recognition**.

	Facial recognition (✓)	Password (✓)	Voice recognition (✓)
This method cannot access the smartphone unintentionally			
The accuracy of this method can be affected by noise			
This method requires the use of the smartphone's camera			
This method could be compromised by shoulder surfing			

[4]

	Facial recognition	Password	Voice Recognition
This method cannot access the smartphone unintentionally		✓	
The accuracy of this method can be affected by noise			✓
This method requires the use of the smartphone's camera	✓		
This method could be compromised by shoulder surfing		✓	

- 11 Many young people play online games. One problem with online gaming is that it is easy to give out personal details accidentally. Some users create weak passwords which could lead to their personal details being accessed.

(a) Name **two** pieces of personal data that could be accessed.

- 1
- 2

11(a)

Two from for example:

Email address
Real names
DOB
Home address
Contact phone number
Membership card number
Location data
Internet Protocol (IP) address
Picture of yourself
Gender

[2]

(b) Write down **three** rules that should be applied **when setting a strong password**.

11(b)

Two from:

Password should not relate to personal details
Should be a long password
Not previously used password
Should not include repeating/obvious patterns/predictable words
Password should be strong

[3]

(c) Describe safety measures that should be taken by gamers to ensure their data is safe other than using strong passwords.

11(c)

Six from:

Use anti-spyware/up to date antivirus software
Play the games with the firewall operational
Play only with authorised versions of games which you have purchased from the correct sources and for which you have a licence
Download/buy files and new software from reputable sources
Do not forget to delete your account details when you are not playing again
Keep the game software up to date.
When disposing of your gaming device ensure all of your personal information has been deleted.
Choose a username that does not reveal any personal information
Be aware of criminals buying or selling 'property' that exists inside a computer game, in the real world

[6]

4 Write down the most appropriate network device to match the following statements. Your answers should be different in each part.

(a) This network device checks the data packet arriving from one computer and sends the data to a specific computer.

..... [1]

(b) This network device sends the data packet from a computer to all devices connected to it.

..... [1]

(c) This network device connects a LAN to the internet.

..... [1]

(d) This network device connects one LAN to another with the same protocol.

..... [1]

Question	
4(a)	Switch
4(b)	Hub
4(c)	Router
4(d)	Bridge

- 5 The Internet of Things (IoT) is a computer network which allows users to control household devices remotely. It has many benefits, for example a user can turn on the house central heating via a smartphone. The IoT can use WiFi and Bluetooth which can cause problems regarding data security.

Describe the methods that could be taken to protect the user's data.

Question	Answer
5	Six from: - Change default name/usernames and passwords on the router - Change the default privacy//use a strong privacy setting - Disable features not in use - Use strong WiFi encryption - Separate the IoT from the home WiFi account - Keep software/hardware up to date - Avoid public WiFi networks - Ensure firewall is operational - Use anti-spyware/up to date anti-virus - Use strong passwords - Use unique passwords for each device - Change passwords regularly

Activate Windows
Go to Settings to activate Windows.

9 in

[6]

- 5 Keeping data secure is very important in any computer system. Many organisations use encryption when sending data.

(a) Describe the term encryption.

5(a)	Four from: Scrambling of data Changes the data into a form that is not understandable Requires a decryption key/encryption key to decode Encrypted using an encryption key/code Changes plain text into cypher text
------	---

[4]

- (b) Another way of protecting data in a computer system is to use passwords.

Identify methods to prevent hackers from gaining knowledge of someone's password.

5(b)	Four from: Use anti-spyware to prevent key logging Change passwords regularly//Do not repeat the same password Use a different password for each system Avoid common/predictable patterns as they are easier to guess Use longer passwords as they are harder to guess Use strong passwords Use two-factor authentication so that hackers need both parts Use a dropdown list for password entry Use a biometric password Do not use passwords that directly link to the user Do not allow webpages/device to remember the password
------	---

[4]

- 2 Tick (✓) whether the following statements refer to the **internet** or an **intranet**.

	internet (✓)	intranet (✓)
This type of network is public		
This is the more secure type of network		
This type of network is global		
This type of network is more likely to be monitored		

[2]

	internet	intranet
This type of network is public	✓	
This is the more secure type of network		✓
This type of network is global	✓	
This type of network is more likely to be monitored		✓

- 7 A company is considering developing a laptop computer that only uses **biometric** methods to authorise user access to this laptop computer. The developers are considering using facial recognition.

- (a) Explain, giving other examples, why **biometric** methods are considered to be a more secure way of authorising access to the laptop computer rather than typing in passwords.

Max five from:

The biometric data is unique to the user

Biometric data is always with you

Passwords can be copied/forgotten/guessed/cracked

Difficult to copy/forgo biometric data

Eliminates shoulder surfing

- Prevents key logging software

Max **two** from, e.g.:

fingerprints, Retina scans, Voice recognition, Iris scans, Handprint

[6]

- (b) Describe the disadvantages of using facial recognition to authorise access to the laptop computer.

Four from:

If the user wears glasses/grows a beard/damages the face/aging effects, then the laptop computer may not read the face correctly
Position of the face and distance from the screen is important
Possible concerns about personal liberty infringement/intrusive
Equipment/set up is more expensive
The system has to be set up before use and this can take more time than setting up a password
May activate accidentally
Once set up it is difficult to reset

[4]

Feb/March 2020/12

- 12 When people use the internet for shopping and banking it is essential that they use a website that has a secure server.

- (a) Describe the features of a web page that identify it as using a secure server.

Four from:

The web address should start with https
There is a closed/green padlock
When the padlock is clicked a message will state the connection is encrypted/secure
The web page has a digital/SSL certificate
The green padlock shows it has an Extended Validation (EV) certificate/CA certificate

[4]

- (b) As more and more people use smartphones as computer systems, security can be breached. Smishing and pharming are ways in which data can be compromised on a smartphone.

Compare and contrast smishing and pharming.

Max four from:

Differences

Pharming downloads a file/program/malicious code onto the user's smartphone

When the user enters a correct website the software redirects them to the fake website

When the user enters his password/personal data it is sent to the hacker's computer

Pharming can be stopped by not downloading .exe, .bat files

In Smishing the user receives a text message from an unknown person

In Smishing the user is asked to phone a number/reply to the text/click on a link

Smishing can be stopped by not replying to unknown text messages

Max three from:

Comparisons

Both are attacks on the smartphone

Both redirect the user to a fake company/person/fake website

Both require the user to enter/divulge personal information

Both obtain/steal personal/your data

[6]

5 Phishing and pharming can affect the security of a user's data.

(a) Compare and contrast phishing and pharming.

5(a)

Max **four** from:

Comparisons

Both use the internet
Both redirect the user to a fake website
Both can lead to fraud and identity theft
Both use websites that look legitimate
Personal data is compromised

Max **four** from:

Differences

With phishing emails are sent to the computer
In pharming malicious code is uploaded to the computer
In phishing clicking a link sends user to a fake website
In pharming the code redirects the user to a fake website without their knowledge
The email in phishing appears to have come from a trusted source
Website in pharming appears to be genuine
In pharming spyware can be used to record key presses

[6]

(b) Smishing is another security issue. Describe the methods which can be used to help prevent smishing.

[6]

Six from:

Avoid clicking on any unknown texts with links/Avoid clicking links in texts from people you do not know.

Do not reply to text messages that have asked you about any of your personal finances/details.

Report any suspicious activity.

Be on the lookout for messages that contain the number 5000 or any number that is not a phone number as this is used by smishers. If the text messages urge you for a quick reply then that is a clear sign of smishing.

Do extensive research before replying to any message. There are plenty websites that allow anyone to run searches based on a phone number and see any relatable information about whether or not a number is legitimate.

Never call back a phone number that was associated with the text from an unknown source.

If the message states 'Dear user, congratulations, you have won.' It is a clear sign for smishing.

Check the time when the unknown message was sent.

If the text message was sent at an unusual time, then that is another sign of smishing.

- 6 Tick (✓) which of the following statements apply to LANs, which apply to WANs and which apply to WLANs.

	LAN (✓)	WAN (✓)	WLAN (✓)
The internet is an example of this type of network			
This type of network is found in a building but is connected without cables			
This type of network is used to transmit data between Europe and North America			
This type of network is found in a building but is connected with cables			

[4]

	LAN	WAN	WLAN
The internet is an example of this type of network		✓	
This type of network is found in a building but is connected without cables			✓
This type of network is used to transmit data between Europe and North America		✓	
This type of network is found in a building but is connected with cables	✓		

- 8 Following recent data protection issues the personal data we allow organisations to share about us may be reduced.

(a) Discuss the benefits and drawbacks of trying to keep personal data confidential and secure.

Max **five** from:

Benefits

If personal data is kept confidential then identity theft is reduced

It helps prevent fraud

Reduces the chance of users suffering harm from criminals, as less data for criminals to attempt to access

Protects sensitive data

... examples:

racial/ethnic/political/religious/membership of trade unions/health/criminal activity

Protects the data being used for purposes other than what it was collected for

Max **five** from:

Drawbacks

Difficult to enforce as users give information freely

Most users do not read the terms and conditions before agreeing to them, therefore allowing data to be used

Data protection is local but networks are global so laws are difficult to enforce

Younger people do not mind that data is passed on therefore the law could not be enforced for all

[6]

(b) The sharing of email addresses can lead to spam.

Explain what is meant by the term spam email.

[2]

8(b)

Two from:

Electronic Junk email/unsolicited emails/sent to a person without requesting it

Deliberate filling up of a user's inbox/many emails sent at once

Emails sent to everyone on the mailing list

(c) Explain why spam email should be prevented.

8(c)	Four from: Takes time to deal with it Can slow down networks Can fill up the receiver's mail box and therefore fill up hard disk/leads to denial of service Can increase the time needed to run your email software Could contain a virus
------	---

[4]

11 Most data protection acts include the principle that data should be accurate and, where necessary, kept up to date.

List four other principles of a typical data protection act.

1.....

Four from: Data should be fairly and lawfully processed Data should only be processed for the stated purpose Data should be adequate, relevant and not excessive/limited Data should not be kept longer than necessary Data should be processed in accordance with the data subject's rights Data should be kept secure Data should not be transferred to another country unless they have adequate protection. Data should be processed in a transparent manner Data should be collected for specified/explicit/legitimate purposes Data should only be further processed for archive purposes which is compatible with the initial purposes Data kept for archiving should safeguard the rights and freedoms of individuals Explicit consent required for processing sensitive data Parental consent required for processing personal data of children including online services Data subjects are allowed access to their personal data	[4]
---	-----

- 3 Complete the following sentences using the most appropriate network term.

A communication method that wirelessly connects a mobile phone to a car, to allow hands-free use of the mobile phone is

A network device that allows a stand-alone computer, using a dial-up analogue connection, to connect to an ISP is a

A network device that allows a LAN to connect to the internet is a

A wireless communication method used in WLAN is

3	Bluetooth Modem Router WiFi
---	--------------------------------------

[4]

- 6 Smishing and phishing can affect the security of a user's data.

(a) Compare and contrast smishing and phishing.

Max **five** from:

Comparisons

Both use communication devices to collect personal data
Both require the user to communicate quickly
Both require the user to enter personal data
Both use the internet

Max **five** from:

Differences

Smishing: fake text messages/SMS are sent to a phone
Phishing: fake emails are sent to your computer
In smishing the user may be asked to make a phone call
A smishing message is shorter than a phishing email
Easier to spot a smishing message
Smishing messages tend to use 5000 in the message

[6]

(b) Describe the methods which can be used to help prevent phishing.

Six from:

Avoid clicking on any unknown emails with links.

Do not reply to emails that have asked you about any of your personal data.

Report any suspicious activity.

If the email urges you for a quick reply then that is a clear sign of phishing.

Do extensive research before replying to any message, check the domain name/company name

If the email is too good to be true, it possibly is

Check the time when the unknown email was sent.

If the email was sent at an unusual time, then that is another sign of phishing.

If the to: box has a large number of similar emails to yours then it is phishing

If there are several grammar mistakes then it is suspicious

[6]

3 A computer system is connected to a LAN and a WAN.

(a) Explain what a LAN is.

3(a)	Two from: Local Area Network A network within a building/site/small geographical area Network connected by hubs and switches
------	--

[2]

(b) Explain what a WAN is.

3(b)	Wide Area Network This network spans a large geographical area
------	---

[2]

- 13 A school is planning to set up a student work files area. The school is considering allowing students to log in to the student work files area either using the school's intranet or the internet.

Discuss the advantages and disadvantages of using the school's intranet rather than the internet to access this student work files area.

Max seven from:

Advantages

Intranet is policed/controlled by the school
More secure as it needs a password
Intranet private network only contains information that is relevant
Intranet resides behind a firewall therefore data is more secure
Messages can target the correct students
Better bandwidth on the intranet
Access from external people are barred

Max seven from:

Disadvantages

More expensive to set up
More time consuming to set up
More time spent on administration i.e. passwords
May only be viewed within the school
More expensive to maintain
More time consuming to maintain
Will not be able to access school files at home for homework
School would need to provide more computers across the school for students to use to access via the intranet

One mark can be awarded for a reasoned conclusion

[8]

3 A variety of network devices can be used to connect computers together.

(a) Complete the following sentences using the most appropriate network device from the list. Your answers must be different in each case.

an access point

a bridge

a hub

a modem

a network interface card

a switch

A device used to connect a Local Area Network (LAN) to another LAN that uses the same protocol is called

A device used to convert a computer's digital signal to an analogue signal for communication is called

A device that connects a number of other devices and sends data to one specific device only is

An item of hardware in a computer that allows a device to connect to a network is

3(a)

A bridge
A modem
A switch
A network interface card

They are in this order
All answers must be different

[4]

A router is used to connect a LAN to the internet. A message is sent from the router to another router.

(b) Give the name of the part of the router where the IP addresses are stored.

.....
..... [1]

3(b)

Routing table

(c) Explain how the data from a router is routed.

3(c)

Five from:

Data is sent in data packets//uses data packets

Each data packet contains an IP address of the next router

The router reads/checks/inspects the IP address

It checks the IP address against its routing table

Data packet is sent to the router with the IP address

The router uses the IP address to work out the best route/destination computer

The router stores the IP addresses

[5]

6 Data stored in a computer needs to be protected.

(a) Describe what is meant by encryption.

6(a)

Three from:

This is the scrambling of data

Makes the data not understandable/meaningless

This system uses an encryption key to encrypt the data

This system uses an encryption/decryption key to decrypt the data

Protects sensitive data

[3]

5 (a) Compare and contrast an intranet with the internet.

.....

Eight from:

Similarities

Both are communication systems
 Both use internet technologies
 Both use HTML to write the web pages
 Both are information systems
 Both are networks

Differences

An intranet has private access whereas the internet has public access
 An intranet is policed whereas the internet anyone can post material
 The data on an intranet is moderate whereas /the internet is not moderated
 The data access on an intranet is more secure than the internet
 Internet has greater amount of information than intranet
 Intranet is within an organisation whereas the internet is world-wide
 Data in the intranet is more relevant/reliable than information found on the internet

To gain full marks there must be at least one from each section

..... [8]

- (b) When customers **shop online**, they can pay using a debit card.

Describe measures that customers could use to reduce fraud when using a **debit card online**.

8(b)

Five from:

Use up to date anti-spyware/anti-virus to stop phishing and pharming attacks

Using websites that encrypt data when sent/green padlock/use HTTPS/SSL

Ensuring that strong passwords are used

Passwords used for the online shopping account should be different to other passwords you use

Check all transactions carried out with the store/receipts

Type in the web address of the supermarket rather than copy and paste/click on links from an email

Do not buy from untrusted websites/apps

Ensure that the wireless password to the router is strong

Report any phishing attempts

Disable cookies/delete cookies frequently

[5]

- (c) Smishing, pharming and phishing are three security threats that can occur when using the internet for shopping.

Tick (✓) whether the following statements refer to **pharming**, **phishing** or **smishing**.

	pharming (✓)	phishing (✓)	smishing (✓)
Fake emails are sent to target users			
When the user types in a URL they are directed to a fake website			
Malicious code is installed on the user's computer			
Fake text messages are sent to the user's phone			

[4]

	pharming (✓)	phishing (✓)	smishing (✓)
Fake emails are sent to target users		✓	
When the user types in a URL they are directed to a fake website	✓		
Malicious code is installed on the user's computer	✓		
Fake text messages are sent to the user's phone			✓

9 Legal documents can be sent by email, as a file attachment, or by physical fax.

(a) Describe the advantages of using emails compared with physical faxing.

Five from:

The document/attachment loads directly into the computer's memory therefore better security

The computer need not be switched on

The computer receives a soft copy

The fax does one thing at once therefore could be engaged

The document in an email is in digital format

Easier to send to multiple recipients

People are more likely to have email rather than fax

Print quality is better with email

More accessible as it is can be sent/received by many devices

The process of faxing is slower as the document needs to be scanned before sending

With fax the paper could get damaged during scanning/printing

Videos and audio can be sent by email

[5]

A legal document is sent as a file attachment.

(b) Explain the steps that need to be taken to ensure that if a virus is attached to the file, it cannot infect the computer.

Four from:

The system must have installed anti-virus software

The anti-virus software must be up to date/updated regularly//download anti-virus

The email/attachment is scanned by the anti-virus software

The email is not opened/attachment downloaded until any virus is removed by the anti-virus software

[4]

A computer virus has been downloaded from an infected file.

(c) Describe **three** effects this could have on a computer.

1	9(c)	Three from:	
.....		Hard disk could be filled up	
2		The computer is performing routine tasks a lot more slowly	
.....		Pop-ups keep appearing on the screen at random.	
3		Random reboots	
.....		Software errors/software/computer doesn't work properly	
		Recycle bin/files not working	
		Computer hanging	
		Anti-virus stops working	
		System lock-up	
		Files are being deleted	[3]

- 8 A company has offices in London and Beijing. The managers of the company meet in person, at a face-to-face meeting twice a year in their conference rooms either in the London office or the Beijing office. The cost of air travel has caused the managers to consider using video-conferencing rather than holding face-to-face meetings.

- (a) Discuss the benefits and drawbacks to the company of using video-conferencing. You should ignore the cost of air travel in your argument.

[8]

- (b) Apart from a keyboard and a mouse, identify **three** other hardware devices which must be present for a video-conference to take place.

- 3 [3]

Question	Answer	Marks
8(a)	Max five from Benefits Can call meetings at short notice Employees save time travelling No hotel expenses Safer as participants do not have to travel No need to transport/carry equipment/lots of documents No risk of losing documents in transit Max five from Drawbacks Takes time/costs money to learn new technology/to be trained Difficult to have international meetings because of time differences Initial cost of hardware (and software) Equipment can break down/power cuts can stop conference Poor strength of signal/time lag/lip sync can be a problem/connection can be lost Loss of personal/social contact Legal documents cannot be signed A mark can be awarded for a reasoned conclusion	8
8(b)	Three from Router Web cam Microphone Network card Speakers Large monitor/data projector Codecs	3

5 A desktop computer is already used in an office which does not have an internet connection.

(a) Name **two** items of hardware which would enable the computer to be connected to a local area network.

1

2 [2]

(b) Name **one** item of hardware which would enable the network to be connected to the internet.

..... [1]

(c) Tick whether these statements are true or false.

	true	false
Booking a theatre ticket is an example of online processing.		
Producing utility bills is an example of online processing.		
An internet browser is used by web designers to test web pages.		
An internet browser uses an inference engine.		

[2]

5 (a) Any two from:

- hub
- switch
- NIC
- bridge

(b) router/modem

[1]

(c)

	true	false
Booking a theatre ticket is an example of online processing.	✓	
Producing utility bills is an example of online processing.		✓
An internet browser is used by web designers to test web pages.	✓	
An internet browser uses an inference engine.		✓

4 correct answers – 2 marks

2 or 3 correct – 1 mark

1 correct – 0 marks

[2]

- 1 For each of the different statements, tick the most appropriate network.

	WAN (✓)	WLAN (✓)	LAN (✓)
A cabled network used in one building.			
The internet is an example of this type of network.			
Several networks connected together using a router.			
This network is connected without cables.			

[4]

Question	Answer				Marks
1		WAN (✓)	WLAN (✓)	LAN (✓)	4
	A cabled network used in one building.			✓	
	The internet is an example of this type of network.	✓			
	Several networks connected together using a router.	✓			
	This network is connected without cables.		✓		

- 4 A router is used in networks.

Describe how a router routes data packets.

.....

.....

.....

.....

.....

.....

..... [3]

Question	Answer	Marks
4	Three from: Inspects the data packets received Checks the destination IP address... ...using the stored routing table It uses a routing table which lists all the different routes to other networks Uses the IP address to work out the best route... ...sends the data packet to the next router	3

12 The internet a

(a) Explain v

(b) Explain what is meant by an intranet.

.....

.....

.....

.....

.....

.....

[3]

May/June2019/11

(c) Describe the advantages of using the internet to research information for a technical project.

- (b)** Bluetooth is an alternative method of allowing computers to communicate with other computers and devices.

Describe how a tablet computer can connect to a printer using Bluetooth.

- (a)** Compare and contrast the use of a hub and a switch.

[4]

Question	Answer	Marks
7(a)	Four from: Similarities A hub and a switch both are used to connect devices to form a LAN Both a hub and a switch use data packets A switch is a type of hub Both check/read the data packets Differences In a hub a data packet is broadcast / sent to every computer or device on the LAN whereas in a switch the data packet is sent to a specific computer Security is lower in a hub as all data is broadcast In the switch the destination address is determined/looked up before it is sent whereas in a hub data packets are sent to all A switch uses MAC addresses to locate the destination of the device whereas in a hub MAC addresses are not checked A switch uses a look up table to determine destination this is not required in a hub A switch is capable of more functionality / multiple VLANs but not in a hub To gain full marks both similarities and differences are required	4

7(b)	Four from: The devices automatically connect when they come into range Tablet sends radio signals to the printer / printer receives radio signals from the tablet Tablet receives radio signals from the printer / printer responds using radio signals A handshake takes place Award a mark for mention of use of 4 digit code for access / default code / auto connection with code / some devices don't need code Uses a band of 79 radio frequencies / channels The tablet computer randomly chooses one of the radio frequencies / channels to use If it is being used it chooses another at random until it finds a free one Uses spread-spectrum frequency hopping 2 marks can be awarded for a good description of the handshake.	4
------	---	---

- 3 A media company with branches around the world has recently opened a new branch in London. The branch includes a number of different computer networks. Identify the most appropriate type of network to answer each of the questions.

(a) The type of network used to connect the branch to its head office in New York is called a

..... [1]

(b) The type of cabled network used to connect computers together in one of the offices is called a

..... [1]

Question	Answer	Marks
3(a)	Wide Area Network	1
3(b)	Local Area Network	1

- 12 A motor car company has some designers based in London and some in Beijing. The cost of travel between the two cities is very high, so when they wish to meet to discuss new products they use video-conferencing.

- (a) The designers all have PCs with a keyboard and a mouse in order to take part in video-conferencing.

Name **three** other devices used to input or output data which would be needed to take part in the video-conference.

1

2

3

[3]

- (b) Describe **three** potential problems of the designers using video-conferencing systems rather than meeting in either London or Beijing.

1

.....

2

.....

3

.....

[3]

- (c) Documents that are sent between the two cities are encrypted.

Explain what encryption is and why it is used.

.....

.....

.....

.....

.....

..... [3]

Question	Answer	Marks
12(a)	Three from: Webcam / video camera Speakers / headset / headphones Large monitor / television / data projector Microphone	3

Question	Answer	Marks
12(b)	Three from: Time lag / lip sync caused by the image not being synchronised with the sound Poor picture quality caused by the speed of connection / quality of the hardware More likely to have poorer sound quality caused by the quality of the hardware / connection Confidential material about the new cars may have to be signed / viewed in person The new car may have to be viewed in person Hardware breakdown stops the conference taking place Communication breakdown stops the conference taking place Different time zones will mean the conference has to take place at inconvenient times	3

12(c)	Max two marks from: Scrambling / encoding of data / convert plain text to cypher text Uses encryption software / key to encrypt it Requires a decryption / encryption key / software to decrypt Max two marks from: Meaningless to the hacker Secures data being transferred from computer to computer Protects sensitive data To gain full marks the answer must include both explanation and use	3
-------	--	----------

- 3 Tick the most appropriate type of network for each description.

	LAN (✓)	WAN (✓)	WLAN (✓)
A cabled network in a building.			
A network that can use satellites.			
A network that connects countries.			
A network that uses wireless technology.			

[4]

Question	Answer				Marks
3		LAN (✓)	WAN (✓)	WLAN (✓)	4
	A cabled network in a building.	✓			
	A network that can use satellites.		✓		
	A network that connects countries.		✓		
	A network that uses wireless technology.			✓	

- 12 A company used to have video-conferences with its suppliers, but due to technical problems they are considering running an audio-conference.

Describe the advantages of using audio-conferencing rather than video-conferencing.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Question	Answer	Marks
12	Four from: Audio-conferencing does not have time lag Lip sync errors cannot occur on audio-conferencing In video-conferencing sound quality can be poorer Video-conferencing needs expensive / extra hardware Video-conferencing requires the internet In video-conferencing there is a lack of stability of the system / communication as there is more chance of it disconnecting / hanging In audio-conferencing only voice is sent / received in video-conferencing sounds and vision is sent / received this can cause issues	4

May/June 2018/13

13 An organisation is planning to train all staff online rather than by video-conference.

(a) Describe what is meant by a web-conference.

.....

.....

.....

.....

.....

.....

[3]

(b) Explain the differences between video- and web-conferencing.

.....

.....

.....

.....

.....

.....

[3]

Question	Answer	Marks
16	Six from: WLAN is wireless therefore devices can be easily added/changed WLAN is wireless so relatively easy to connect to portable devices WLAN uses radio signals/wireless technology/wifi WLAN uses (wireless) access points/wireless node WLAN has a limited range WLAN signal strength can diminish the further away from the access point WLAN signals affected by walls/obstacles WLAN is more prone to hacking than cabled systems WLAN susceptible to interference from another radio signals	6

- 18** As more and more people use the internet at home to carry out transactions they are at risk of downloading software that could corrupt files or copy credit card details.

Discuss the effectiveness, or otherwise, of using different methods of increasing the security of a home computer.

[8]

Question	Answer	Marks
18	To be marked as a level of response: Level 3 (7–8 marks): To gain a level 3 there must be a <u>reasoned conclusion</u> and a <u>justification</u> and must have <u>both sides</u> of the argument Award a mark for justification of the points raised Award a mark for a reasoned conclusion Level 2 (4–6 marks): For level 2 there must be more than three statements; which cover <u>both sides</u> of the argument and achieved all of Level 1 Level 1 (1–3 marks): For level 1 there must be statements up to max three Level 0 (0 marks): Response with no valid content Examples of answers written below Answers may make reference to, for example: <u>Up to date antivirus</u> software needed/scan files/attachments when downloading/install anti-virus software Computer/emails/uploads need to be scanned regularly <u>Firewall required</u> to stop attacks from suspicious computers Has a blocked website list/has a white list A firewall can be looked through therefore two are better than one Stops fraudulent sites attacking the computer <u>Up to date anti-spyware software can be used</u> <u>Use of strong passwords</u> Changing passwords from default passwords <u>Regularly changing passwords</u> <u>Using different passwords for sites</u> Not using a computer/mobile device in a public area Data on the cloud needs to be encrypted and have a strong password Good antivirus can be expensive to purchase Free antivirus software may not be up to date This can cause memory issues in the computer as updates use memory Antivirus needs to be in operation at all times slowing the operation of the computer <u>Some legitimate software can be blocked by the firewall</u> Spyware sends a user to a fraudulent website and records key logs User does not know that the spyware has been uploaded Popups are stopped by anti-spyware, pop-under are not Too complex a password can be easily forgotten Password files may not be backed up Hackers can breach most passwords	8

- 2 Tick whether the following activities relate to the use of **email**, **electronic fax** or **both**.

	email (✓)	electronic fax (✓)	both (✓)
Can be used as a legal document			
Allows video attachments to be added to the message			
Requires a telephone number			
Messages can be sent to a computer			

[4]

Question	Answer				Marks
2	1 mark for each correct row.				4
		email (✓)	electronic fax (✓)	both (✓)	
	Can be used as a legal document			✓	
	Allows video attachments to be added to the message	✓			
	Requires a telephone number		✓		
	Message can be sent to a computer			✓	

- 11 Tick whether the following activities relate to an intranet, the internet or both.

	intranet (✓)	internet (✓)	both (✓)
This is a restricted access system.			
This system can use email for communication.			
This is a public system.			

[3]

Question	Answer				Marks
11	1 mark for each correct row.				3
		intranet (✓)	internet (✓)	both (✓)	
	This is a restricted access system	✓			
	This system can use email for communication			✓	
	This is a public system		✓		

- 3 Pupils working on a science project use the school network to store and retrieve files. They log onto the network using individual passwords.

(a) Abisha is using his name as his password.

Give **two** reasons why this is not a good idea.

1

.....

2

.....

[2]

- (b) Give **two** other pieces of advice the teacher should give regarding the management of passwords.

1

.....

2

.....

[2]

Question	Answer	Marks
3(a)	Two from: Password is too short Easy to guess Personal information used in the password Doesn't contain a mixture of upper case, lowercase, numbers or symbols	2
3(b)	Two from: Change passwords regularly Use a strong password Do not write the password down Do not tell anyone else the password Must be easy for the user to remember	2

- 9 Describe how a router forwards data packets from one network to another.

.....

.....

.....

.....

.....

..... [3]

Question	Answer	Marks
9	Three from: The router reads the data packet destination address Looks up the paths to get to that address Compares to its routing table to get to the destination address Checks how busy the paths are Sends the packet on the least congested/quickest path	3

14 Emails are a very popular way of communicating.

Tick **four** statements, relating to emails that are true.

	Tick (✓)
Emails can only be sent from a computer.	
An email can be used to send a message to many people at the same time.	
It always takes a long time for an email to arrive.	
All emails must have an attachment.	
Attachments may have to be compressed to be sent via email.	
Emails can be used to send legal documents.	
Every email has to be printed out.	
In order to send an email back to the sender without retyping the address, reply is used.	
Email means extended mail.	
Email attachments are always checked for viruses before they are sent.	

[4]

Question	Answer		Marks
14		Tick (✓)	4
	Emails can only be sent from a computer.		
	An email can be used to send a message to many people at the same time.	✓	
	It always takes a long time for an email to arrive.		
	All emails must have an attachment.		
	Attachments may have to be compressed to be sent via email.	✓	
	Emails can be used to send legal documents.	✓	
	Every email has to be printed out.		
	In order to send an email back to the sender without retyping the address; reply is used.	✓	
	Email means extended mail.		
	Email attachments are always checked for viruses before they are sent.		

2 Tick the most appropriate device matching the statements.

	Hub (✓)	Switch (✓)	Router (✓)
Connects a LAN to a WAN.			
Broadcasts data to all connected devices.			
Connects two LANs.			
Selects which device to send data to.			

[4]

Question	Answer				Marks
2		Hub (✓)	Switch (✓)	Router (✓)	4
	Connects a LAN to a WAN			✓	
	Broadcasts data to all connected devices	✓			
	Connects two LANs			✓	
	Selects which device to send data to		✓		

9 A London based radio company is planning to cover the Commonwealth Games in Australia. Their staff will hold video-conferences with the games organisers.

(a) Describe what is meant by video-conferencing.

[4]

- (b) Describe **four** disadvantages to the radio company of using video-conferencing rather than face-to-face meetings in this scenario. A face-to-face meeting is when the staff attending the meeting are all together physically in one room.

1

.....

2

.....

3

.....

4

.....

[4]

Question	Answer	Marks
9(a)	Four from: Video-conferencing is the use of telecommunication technologies It uses an internet connection It allows people in two or more locations to communicate Allows long distance communication Using simultaneous two-way communication Uses video and audio transmissions/streaming	4
9(b)	Four from: Video conferencing equipment can be expensive to buy The communication link could be expensive to set up and operate/satellite/internet link needed Lip sync/lagging can be an issue If the hardware breaks down the conference fails // poor connection conference fails Time zones as the games are in Australia and company is based in London Needs organising as to who's turn it is next // not talk at the same time Any sound in the conference can be picked up by microphones Open microphones can distort the sound	4

- 11 Security of data is a major concern for people using computer systems. One method of protecting the data is to use a strong password. This is an authentication technique.

(a) Name **three** other authentication techniques that could be used.

1

2

3

[3]

- (b) A typical data protection act tries to protect the data of living individuals. One way of achieving this is by keeping data secure.

Identify **four** other principles of a typical data protection act.

1

2

3

4

[4]

Question	Answer	Marks
11(a)	Three from: Biometrics e.g. iris scan, face scan, retina scan Physical Tokens/TAN Two-factor/two-step/multi-factor authentication Magnetic stripes ID cards Memorable information // security question	3

Question	Answer	Marks
11(b)	Four from: Data must be processed fairly and lawfully Data must be used for the purpose it was collected for Data must be adequate and relevant to the purpose it was collected for Data must be kept up to date and accurate Data must not be kept longer than necessary Data must be processed within the rights of data subjects Data must not be moved to any other country that is not protected by a data protection act Must allow access to data	4

- 3 Tick whether the following statements about networks are **true** or **false**.

	true (✓)	false (✓)
A hub is a central computer.		
A router connects a LAN to the internet.		
A bridge connects a WAN to a LAN.		
A switch connects computers in a network.		

[2]

Question	Answer	Marks															
3	<table> <tr> <th></th><th>true (✓)</th><th>false (✓)</th></tr> <tr> <td>A hub is a central computer.</td><td></td><td>✓</td></tr> <tr> <td>A router connects a LAN to the Internet.</td><td>✓</td><td></td></tr> <tr> <td>A bridge connects a WAN to a LAN.</td><td></td><td>✓</td></tr> <tr> <td>A switch connects computers in a network.</td><td>✓</td><td></td></tr> </table> 4 correct answers – 2 marks 2 or 3 correct answers – 1 mark 0 or 1 correct – 0 marks		true (✓)	false (✓)	A hub is a central computer.		✓	A router connects a LAN to the Internet.	✓		A bridge connects a WAN to a LAN.		✓	A switch connects computers in a network.	✓		2
	true (✓)	false (✓)															
A hub is a central computer.		✓															
A router connects a LAN to the Internet.	✓																
A bridge connects a WAN to a LAN.		✓															
A switch connects computers in a network.	✓																

7 Explain the purpose of an intranet.

[3]

Question	Answer	Marks
7	Any three from: Private/confidential data stays within a company/organisation/school To store relevant/up to date information about the organisation available to workers To facilitate communication/collaboration between workers/students/teachers Make it easier to access company specific applications. Restrict access to certain sites of the internet/regulate access to the internet Restrict access for certain employees	3

18 Many devices in the home can now connect directly to the internet.

Smartphones and computers used in the home can connect to the internet.

Discuss the benefits and drawbacks of using other devices that connect to the internet in the home.

[8]

Question	Answer	Marks
18	To be marked as a level of response: Level 3 (7–8 marks): Candidates will address both aspects of the question and discuss/consider different benefits and drawbacks. The issues raised will be justified. There will be a reasoned conclusion. The information will be relevant, clear, organised and presented in a structured and coherent format. Level 2 (4–6 marks): Candidates will address both aspects of the question and discuss/consider different benefits and drawbacks although development of some of the points will be limited to one side of the argument. There will be a conclusion. For the most part the information will be relevant and presented in a structured and coherent format. Level 1 (1–3 marks): Candidates may only address one side of the argument, and give basic benefits and drawbacks. Answers may be simplistic with little or no relevance. Level 0 (0 marks) Response with no valid content.	8
	<i>Answers may make reference to e.g.:</i> benefits of smart technology Smart technology devices can be used to turn on devices remotely.... ...if you are coming home you can turn on cookers ...you can come home to a warm house by turning on central heating Smart fridges can analyse food taken out of fridge so you can get a healthy diet ...also re-orders stock automatically Smart washing machines stop washes if incorrect materials present in the wash if new materials are produced then the washing machine can analyse the materials and tell you the correct dose of powder if you want to watch up to date films then the TV can connect to the internet drawbacks to smart technology Over reliance on technology; if the device breaks down then it is difficult to use alternatives If the device breaks it is more difficult to repair Personal data can easily be hacked through internet connected devices ...personal data easily found ...wifi key easily found If controller/smart phone lost/stolen the devices may not work in the home New technology is expensive Too many gadgets reliance on internet connection.	

8 A company is setting up a video-conference.

(a) Name **three** computer hardware devices they would need.

- 1.....
-
- 2.....
-
- 3.....
-

[3]

(b) The company could have set up a web-conference rather than a video-conference.

Describe what is meant by a web-conference.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Question	Answer	Marks
8(a)	Any three from: – Video/digital camera/webcam – Microphone – Keyboard – Large screen/monitor/data projector – Remote control – Speakers/headphones	3
8(b)	Any four from: – A conference held over the internet using TCP/IP connections – Examples webinars/webcasts/VOIP – Can be point to point (VOIP) or multicast – Allows text communication – Allows voice communication – Allows video communication – Uses a web browser	4

2 Tick whether the following statements are **true** or **false**.

	true (✓)	false (✓)
A router is required to allow a LAN to connect to the internet		
A web browser displays a web page		
An intranet has restricted access		
ISP means Internal Service Provider		

[2]

Question	Answer			Marks
2		true (✓)	false (✓)	2
	A router is required to allow a LAN to connect to the internet	✓		
	A web browser displays a web page	✓		
	An intranet has restricted access	✓		
	ISP means Internal Service Provider		✓	
4 correct answers 2 marks 2 or 3 correct answers 1 mark 1 or 0 answers no marks				

16 Bluetooth and WiFi can connect devices together.

Discuss the advantages and disadvantages of using Bluetooth or WiFi.

A blue five-pointed star is centered on a white background. The background features horizontal dotted lines, similar to those found on primary writing paper. The star is a solid blue color with a thin black outline.

[8]

Question	Answer	Marks
16	To be marked as a level of response: Level 3 (7–8 marks): Candidates will give reasoned advantages and disadvantages. There will be a reasoned conclusion. They will relate the answer to both Bluetooth and WiFi. The information will be relevant, clear, organised and presented in a structured and coherent format. Level 2 (4–6 marks): Candidates will expand on advantages/disadvantages relating the answer to both Bluetooth and WiFi. Some of the points may be one sided. There may be a conclusion. For the most part the information will be relevant and presented in a structured and coherent format. Level 1 (1–3 marks): Candidates only list an advantage/disadvantage. Candidates only refer to Bluetooth or WiFi. Answers may be simplistic with little or no relevance. Level 0 (0 marks) Response with no valid content	8
	<i>Answers may make reference to, for example:</i> Bluetooth is more useful when transferring data between two devices whereas WiFi is more suited to full scale networks Bluetooth does not buffer The devices connected via Bluetooth should be near to each other whereas WiFi has a larger range Less data tends to be transferred with Bluetooth Bluetooth uses a lower bandwidth Eg sound data (phones) or file transfer (computers) WiFi has better security than Bluetooth Cost of Bluetooth is cheaper than WiFi Bluetooth has a shorter password Bluetooth devices are battery operated which needs to be replaced or recharged but WiFi do not necessarily use batteries To connect to the network need a Bluetooth adapter but WiFi needs a WiFi adapter, a wireless router and a wireless access point Bluetooth range is approx. 20m whereas WiFi is approx. 100m Bluetooth is simple to use and setup but WiFi is more complex Bluetooth/Wifi are both wireless can't fall over the wires/move around the house Saves money (on cables) as both do not require wires Bluetooth is more energy efficient than wifi in some cases 3% of the energy	

2 Complete each sentence below using the most appropriate item from the list.

a bridge a hub a keyboard a microphone a mouse
a printer a router a scanner a sensor a switch

- (a) A network device which broadcasts data passing through it is called
[1]
- (b) A network device that connects a LAN with the internet is called
[1]
- (c) A network device that connects a LAN to another LAN using the same protocol is called
[1]
- (d) A network device that learns which devices are connected to which ports is called
[1]

Question	Answer	Marks
2(a)	a hub	1
2(b)	a router	1
2(c)	a bridge	1
2(d)	a switch	1

6 Describe the differences between the internet and an intranet.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

Question	Answer	Marks
6	Any four from: The internet is a public network whereas an intranet is a private network/restricted access An intranet is policed or managed whereas the internet has no rules The data on an intranet is moderated whereas the internet is not moderated An intranet is more secure than the internet Internet has greater amount of information than an intranet	4

- 8 Computers can be subject to hacking.

Explain what is meant by hacking and the effect it can have on the data or the computer.

.....

.....

.....

.....

..... [2]



Question	Answer	Marks
8	<u>Unauthorised access to computer systems</u> 1 mark Any one from: 1 mark Data on the computer system can be corrupted, edited, deleted, viruses added Data on the computer system can be copied/stolen/misused Data on the computer system can be accessed/read	2

- 9 A technician has told Moira that her computer might have a virus.

Describe **three** reasons why the technician might think this.

1

.....

2

.....

3

.....

[3]

Question	Answer	Marks
9	Any three from: She has not much space left on her hard disk even though she had previously Her computer is performing routine tasks a lot more slowly. Pop ups keep appearing on her screen at random. Her friends keep telling her that emails about strange things are being sent by her Random reboots Software errors Recycle bin not working Computer hanging Anti-virus stops working	3

10 A new employee of a company is required to log on to the company's computer system and will need to create a suitable password.

(a) Give **three** rules that the company would have in place, to help the employee choose a password.

1

2

3

[3]

(b) He changes his password to R4hB5&Fg1

Explain why he has chosen this password.

.....

.....

.....

.....

.....

[3]

- (c)** When a password is changed it is verified.

Identify the type of verification which is carried out.

.....[1]

Question	Answer	Marks
10(a)	Any three from: Use of strong passwords Do not use pet names Do not use personal information Do not use birthdates Do not use a password they have used before Do not use company name	3
10(b)	Any three from: Conforms to company rules Hard to guess Stop shoulder surfing Hard to crack	3
10(c)	Double data entry	1

- 15 Many people think there should be greater control of the use of the internet.

Describe the advantages and disadvantages of policing the internet.



[5]

Question	Answer	Marks
15	Any four from: Advantages Helps to prevent illegal materials from being shown Helps to prevent pornography (for minors) Helps to prevent discussion groups that incite hatred, racism, etc Ensure copyright laws are enforced Remove the Dark Net Safeguards the vulnerable Any four from: Disadvantages Unsuitable material is easily available in other ways The rise of mobile communications lessens the impact of the internet Freedom of speech is restricted Very difficult to enforce as the internet spans continents Maximum 5 marks	5

- 1 Tick whether the following statements would apply to a **LAN**, a **WAN** or a **WLAN**.

	LAN (✓)	WAN (✓)	WLAN (✓)
The internet is an example of this type of network			
A wired network found within a single building			
This network uses wireless technology			
This network is spread over a large geographical area			

[4]

Question	Answer				Marks
1		LAN (✓)	WAN (✓)	WLAN (✓)	4
	The internet is an example of this type of network		✓		
	A wired network found within a single building	✓			
	This network uses wireless technology			✓	
	This network is spread over a large geographical area		✓		

5 (a) Describe the processes a web browser has to go through to load a web page.

.....[2]

An organisation is creating a company website and is deciding whether to make it available on the internet or create a company intranet. The website will contain confidential information.

(b) Explain why they should use an intranet rather than the internet.

[6]

Question	Answer	Marks
5(a)	Any two from: Request made by entering a web address/URL or clicking on a link Retrieves code from the server/resources downloaded Interprets the code as a web page/uses page resources to build page The web page is displayed/renders page to view	2
5(b)	Any six from: An intranet has restricted access/private network therefore public cannot gain access/the internet is public access The intranet is protected by passwords/More chances of hacking on the internet The intranet has reduced unauthorised access to material The material may be sensitive on the intranet Intranet only used in the organisation/internet is world wide An intranet will only contain relevant information for the company The company has more control of data on an intranet An intranet is policed or managed/the internet has no rules The data on an intranet is moderated/the internet is not moderated	6

February/March

6 Use examples to describe the difference between a weak password and a strong password.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

6 Good example of a strong password – (1 mark)

Good example of a weak password – (1 mark)

Any **two** from:

Weak password is easy to guess/strong password is difficult to guess

Weak passwords consist of all letters or all digits or all digits with one letter/all letters with one digit

Weak password is one that has few characters

Strong password uses mixture of upper case letters, digits and punctuation marks

Weak password has repeated characters

Weak password has characters next to each other on a QWERTY keyboard

Weak passwords are ones that are similar to the user name

Weak passwords are passwords that relate to the user

[4]

8 Describe how data is sent from a networked computer to a computer on a different network.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

8 Any **four** from:

Router

The network software in the original computer determines that the destination computer is on a different network...

...using IP addresses

Data is sent to the switch/hub and then passed to router

the router then passes it to the router of the other network

that router passes it to the switch/hub...

...to the destination computer

Bridge

The computer sends the data to the bridge via the hub/switch...

...using MAC addresses

if the protocol/MAC address is same, bridge passes data to the other network

Bridge passes it to the switch/hub...

...to destination computer with correct MAC address

[4]

9 (a) Describe **two** similarities between physical faxing and electronic faxing.

1

.....

2

.....

[2]

(b) Describe **three** differences between physical faxing and electronic faxing.

1

.....

2

.....

3

.....

[3]

9 (a) Any two from:

Both involve you being assigned a fax (phone) number
Both can involve use of a hard copy original
Both require the use of a phone line

[2]

(b) Any three from:

Physical faxing requires purchase of a fax machine/electronic faxing requires use/purchase of a computer/scanner
Physical faxes can be picked up by anyone/electronic faxing goes straight to your email address
Physical faxes – phone-line could be busy/engaged – electronic faxes – phone line never registers as busy/engaged
Physical faxes – must have paper and can get paper jams
Electronic fax is quicker to arrive than physical

[3]

May/June 2016/11

5 Most data protection acts include the principle that data should be stored securely.

List **four** other principles of a typical data protection act.

- 1
- 2
- 3
- 4

[4]

5 Four from:

Information must be processed fairly and lawfully
Information collected must be processed for limited purposes
Information collected must be adequate, relevant and not excessive
Information collected must be accurate and up to date
Information must not be held for longer than is necessary
Information must be processed in accordance with the individual's rights
Information should not be transferred outside the area of the Act unless adequate levels of protection exist.

[4]

[illegible]

[4]

8	Four from: Stores IP addresses IP address is a unique identifier set up by network manager/ISP Can change but should match the network it's on IP address consists of 4 numbers separated by full stops Stores MAC addresses 6 pairs of hexadecimal digits MAC address is usually hard coded by manufacturer, never changes	[4]
---	---	-----

MAC address is usually hard coded by manufacturer, never changes

[4]

[illegible]

[8]

19 To be marked as a level of response:

Level 3 (7–8 marks)

Candidates will describe in detail the reliability and unreliability of at least two methods of preventing unauthorised access to data.

Examples will be given and will be appropriate.

The information will be relevant, clear, organised and presented in a structured and coherent format.

Specialist terms will be used correctly and appropriately.

Level 2 (4–6 marks)

Candidates will describe the reliability or otherwise of methods of preventing unauthorised access to data.

For the most part, the information will be relevant and presented in a structured and coherent format.

Examples will be given and will be mostly appropriate.

Specialist terms will be used appropriately and for the most part correctly.

Level 1 (1–3 marks)

Candidates will identify some of the methods of preventing unauthorised access to data.

Answers may be in the form of a list.

There will be little or no use of specialist terms.

Errors of grammar, punctuation and spelling may be intrusive.

Level 0 [0 marks]

Response with no valid content

Examples of reliability of methods

User id and Password will...

...prevent users who do not know the password from gaining access

...will reject users who try to guess passwords (usually after 3 attempts)

Strong passwords using a mixture of alphabetic, alphanumeric and special characters...

...will be difficult to guess

Biometrics are unique and are almost impossible to duplicate...

...only the user who has those characteristics can access the laptop

Using biometrics means that passwords don't have to be remembered

Laptops can now come with fingerprint scanner

Laptops can have built in retina identification

Examples of lack of reliability

Strong passwords can be difficult to remember

...can be easily forgotten

...can be easily disclosed to any user

Software for retina scan can malfunction

User can have fingerprints affected by injury/cut on a finger

Laptops with this technology tend to cost more

Low cost fingerprint technology tends to be inaccurate

Fingerprints can be copied by expert thieves

[8]

3 Tick whether the following statements would most appropriately apply to the **internet** or an **intranet**.

	internet (✓)	intranet (✓)
A public computer network.		
Is easier to protect sensitive data from hackers.		
The content is more easily controlled.		
Exists within a single organisation only.		

[2]

3

	Internet (✓)	Intranet (✓)
A public computer network	✓	
It is easier to protect sensitive data from hackers		✓
The content is more easily controlled		✓
Exists within a single organisation only		✓

4 correct answers – 2 marks
 2 or 3 correct answers – 1 mark
 1 correct – 0 marks

[2]

- 6 A school's Science department has a network which uses WiFi. A teacher has a laptop computer and whilst in school wishes to connect to this network.

(a) Write down the name of this type of network.

..... [1]

(b) In order to access the network the laptop must be within range of a type of WiFi network device.

(i) Write down the name of this type of device.

..... [1]

(ii) Describe how this device is connected to the network using an ethernet cable.

.....
 [1]

(c) WiFi networks can be limited in their range.

Describe **one** other disadvantage of a WiFi network compared to a cabled network.

.....
 [1]

6 (a) A WLAN [1]

(b) (i) A (Wireless) Access Point/wireless node [1]

(ii) It is connected to a switch/hub [1]

(c) Can get interference from another radio signal/speed of data transmission can be slower
 Other valid answers like physical obstacles/walls [1]

7 (a) Explain what is meant by VOIP.

.....

 [2]

- (b) Name **two** items of computer hardware that are needed in order to make a VOIP phone call from one computer to another.

1

2

[2]

- 7 (a) Two from:

Voice over Internet Protocol

A set of rules that enable people to use the Internet to make telephone calls/talk each other

Sends voice data in packets using IP

[2]

- (b) Two from:

Microphone

Speakers/headphones

Headset

[2]

- 3 Tick whether the following statements are **true** or **false**.

	true (✓)	false (✓)
WLAN stands for Wide Local Area Network.		
A WAN often requires the use of a router.		
The internet is a WAN.		
A LAN usually covers a smaller area than a WAN.		

[2]

3

	TRUE (✓)	FALSE (✓)
WLAN stands for Wide Local Area Network		✓
A WAN often requires the use of a router	✓	
The internet is a WAN	✓	
A LAN usually covers a smaller area than a WAN	✓	

4 correct answers – 2 marks

2 or 3 correct answers – 1 mark

1 correct – 0 marks

[2]

3 Tick whether the following statements are true or false.

	true (✓)	false (✓)
A WAN usually connects a number of LANs.		
A LAN is usually found in one building.		
A WAN is a Wireless Area Network.		
The internet is an example of a WLAN.		

[2]

3

	TRUE (✓)	FALSE (✓)
A WAN usually connects a number of LANs.	✓	
A LAN is usually found on one building.	✓	
A WAN is a Wireless Area Network.		✓
The internet is an example of a WLAN.		✓

4 correct answers – 2 marks
2 or 3 correct answers – 1 mark
0 or 1 correct answer – 0 marks

[Max. 2]

1 There are a number of different network devices associated with computers.

Give the most appropriate type of communication device that fits the following descriptions.

(a) This device connects a LAN to the internet.

.....[1]

(b) This device transmits data to specific computers in a network.

.....[1]

(c) This device broadcasts data to all computers in a network.

.....[1]

(d) This device connects one LAN to another LAN that uses the same protocol.

.....[1]

1	(a) Router/Broadband modem	[1]
	(b) Switch	[1]
	(c) Hub	[1]
	(d) Bridge	[1]

- 5 The directors of a multi-national car company need to communicate with each other. They have decided that the best way to do this is through web-conferencing.

(a) Describe web-conferencing and how it could be used in this scenario.

.....

.....

.....

.....

.....

.....

.....

.....[4]

- (b) The directors wish to have a conference with their managers in other countries and are considering the use of video-conferencing.

Name **three** pieces of hardware that would be needed in order to hold a video-conference.

1

.....

2

.....

3

.....

[3]

(c) Describe **three** drawbacks of setting up and using video-conferencing.

.....[6]

5 (a) Four from:

Real time collaboration/communication

Multiple users are connected to the internet.

See the same screen at all times in their web browsers

Allows: texting, for example of the car details

VOIP, for example, the delegates discussing the car range

Full motion video to show the car being test driven

[4]

(b) **Three** from:

Webcams/cameras

Large monitors/projector

Microphones

Speakers/headphone

[3]

(c) **Three** matched pairs from:

- Power cuts/load shedding...
...can lead to parts of the video-conference shutting down.
- Lip sync problems/time lag...
...this means that the picture is not running at the same speed as the sound.
- Need to consider time zone implications of when to hold the video-conference...
...some participants may be video-conferencing outside of work hours/in the night.
- Initial cost of hardware/software...
...expensive to buy extra hardware and software
- Needs reliable internet access...
...otherwise the video-conference will suffer interference/break down of signal.
- Difficult to stop them all talking at once/manage contributions...
...if more than one person talks at once it is difficult to hear what is said.
- Difficult to manage the video-conference starts on time...
...due to time differences/setting up issues.
- Difficult to keep participants on task...
...they are not in the same room so could be doing other things/not concentrating.
- Increased cost/time of training staff...
...The host of the conference would need training/participants may need training.
- Cost/time of technician/support staff...
...the support staff will need to be on hand in case of problems.
- Document cannot be signed...
...there will be only one copy of the document.
- Physical objects cannot be examined...
...this would need to be set up before the conference which takes time.

[6]

[6]

[illegible]

[6]

8 Six from:

Examples

Advantages

Biometrics are unique...

...so others cannot sign people in/whereas with password/card systems someone else can sign you in.

Fingerprints/eyes are always with you...

...but a password/card can be lost.

Cannot be forgotten passwords can be forgotten

The user has to be present therefore more secure.

Typing in passwords can lead to errors on data entry...

...biometrics tend to be more reliable.

More difficult to forge...

...passwords can be copied/cards can be stolen.

Disadvantages

Personal liberty infringement...

...fingerprints/biometric must be stored.

Method is slower to enter system than passwords.

Equipment and setup is more expensive than producing cards.

The time taken to set up a store of fingerprints/eye prints can be a long time...

...quicker to set up a database of passwords.

If the finger is damaged/dirty it cannot be read passwords can be more reliable.

If dark glasses are worn retina cannot be read passwords are more reliable

If a voice is not understandable on a voice recognition system may not work.

Facial hair can stop facial recognition passwords are more reliable

A mark can be awarded for a reasoned conclusion

If one side of the argument i.e. all advantages/disadvantages then 3 marks max.

If both sides have been addressed but without expansions/differences then 4 marks max

[6]

16 (a) Describe what is meant by an intranet.

.....

.....

.....

.....[2]

(b) Describe what is meant by the internet.

.....

.....

.....

.....[2]

16 (a) Two from:

Internal restricted access network/secure network
Uses same protocols as the internet
Private network
Used within an organisation eg school
Limited resources

[2]

(b) Two from:

Public network/not policed
International network of networks
WAN/wide area network
World wide/global/international
Public information system
web/email/gophers/social networking/video conferencing is part of the internet

[2]

October/November

10 There are several ways in which members of an organisation can organise meetings. These include video-, audio- or web-conferencing.

(a) An organisation is planning an important meeting with the Directors of the three main offices. They are going to set up an audio-conference.

Describe the process of setting up this type of conference.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

- (b) The organisation has a computer network which is connected to the internet. The router contains a firewall.

Explain why a firewall is needed.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

10 (a) Four from:

Agree the date and time of the conference

Either

Contact an audio conferencing provider ...

...for organiser's PIN

...for participant's PIN

Give the PIN to each participant

at the start of the conference organiser types in his PIN

Each participant dials the number and enters the PIN

or

the organiser phones/contact the first participant

the organiser phones/contact subsequent participants

puts them on hold...

...until all participants have been contacted

joins the participants together/joins the conference

[4]

(b) Four from:

It is needed for network security

It controls the incoming and outgoing network traffic.

A firewall establishes a barrier between a trusted, secure internal network and the Internet

It logs incoming and outgoing traffic

It stops malicious traffic

It prevents computers connecting to unwanted sites

It prevents untrusted/unrecognised computers accessing data on the system

- 15 (a) A router is connected to a LAN.

Describe the function of a router.

.....

.....

.....

.....[2]



- (b) Different methods can be used to connect computers to networks.

Describe **two** advantages for each of the following methods.

WiFi

.....

.....

.....[2]



Satellite

.....

.....

.....[2]

3G/4G

.....

.....

.....[2]

- 15 (a) Two from:

Routers inspect/read the IP address of the data packets sent to it
 Sends the data packet to a switch with that IP address
 It may use the MAC address of the switch to do that by converting the IP to a MAC using ARP...
 ...until it finds the corresponding switch

[2]

(b) WiFi

Any **two** from:

- Can connect to the internet from any room in a house
- No cables
- Can be used on multiple devices

[2]

Satellite

Any **two** from:

- This is broadband
- Useful for general use of the internet
(Connects via satellites so) better coverage
- Can be used almost anywhere in the world

[2]

3G/4G

Any **two** from:

- Used in mobile devices laptops/tablets/smartphones
- Connects wireless through the 3G or 4G network
- Allows access on the move
- Allows access even if no WiFi available
- Not affected by rain or snow as satellite communication

[2]

February/March 2015/12

- 4 There are a number of different types of area networks.
Tick whether the following statements apply to a WAN, LAN, or a WLAN.

	WAN ✓	LAN ✓	WLAN ✓
This type of network usually requires the use of WiFi			
Usually is cabled and often housed in a single building			
Covers a large area and is often connected using a router			
Requires the use of Wireless Access Points (WAPs)			

[4]

4

This type of network usually requires the use of WiFi			✓
Usually is cabled and often housed in a single building		✓	
Covers a large area and is often connected using a router	✓		
Requires the use of Wireless Access Points (WAPs)			✓

[1]

[1]

[1]

[1]

10 Apart from security aspects, describe the role of a proxy server.



[3]

10 Three from:

- Can act as a web server
- Can act as a buffer between internet and LAN
- Server passes on requests to the internet
- Passes the requested web pages to individual computers
- Can cache/store the webpages
- Subsequent requests for that/those web page(s) are responded to more quickly
- Can be used to monitor internet usage

[3]

15 Many computer networks are connected using cables but some are now connected using wireless technology.

Discuss the advantages and disadvantages of using wireless networks compared to using cabled networks.



[6]

15 Six from:

Advantages

Less likely to trip over cables
Easier to relocate computers
Cheaper as you don't have to buy cables
Adding devices/computers to network is easier

Disadvantages

Limited area of network
Strength of signal is weaker
Physical obstacles can interfere with signal/can cause disconnection

One mark is available for a reasoned conclusion.

[6]

October/November

8 Describe each of the following types of network.

LAN

.....

.....

.....

WAN

.....

.....

.....

WLAN

.....

.....

.....

[6]

8

LAN

Two from:

- LAN is a Local Area Network
- LAN covers a small area/one building
- A school network is a LAN

WAN

Two from:

- WAN is a wide area network
- WAN covers a large geographical area/worldwide
- The Internet is a WAN
- A WAN consists of connected LANs

WLAN

Two from:

- WLAN is a wireless local area network
- Uses wireless technology to transmit data
- Uses Wireless Access Points (WAPs) connected to a wired network

[6]

9 Describe four items which can be provided by an ISP.

October/November

- 1
- 2
- 3
- 4

[4]

9

Four from:

- Access to the internet
- Email address
- Router to connect to the internet
- Domain name to have own website
- Email security software
- Web space to create own website
- Storage space to save data in the cloud
- DNS service to convert URLs to IP addresses
- Firewall to block unauthorised traffic

[4]

11 Explain why encryption is needed when transmitting data.

.....

.....

.....

.....

.....

.....[3]

11 Three from:

To cause data to be scrambled/encoded
 Protects sensitive data...
 ...from being understood if it falls in to the wrong hands
 Only user/computer with key can understand data

[3]

17 Give **three** reasons why some companies have intranets.

1

.....

2

.....

3

.....

[3]

17 **Three from:**

It is possible to prevent employees accessing undesirable websites
 They can ensure that available information is specific to their needs
 It is easier to make sure confidential messages stay within the company
 There is less chance of company computers being hacked

[3]

6 Explain, using examples, what is meant by authentication techniques.

.....

.....

.....

.....

.....

..... [3]

6 Three from:

Is a means of identifying yourself to a system
 Something you know such as PIN/password
 Something belonging to you such as a bank card
 Something unique about you such as a biometric feature

[3]

10 Describe the differences in use between WiFi and Bluetooth.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

10 Four from:

WiFi technology enables local area network...
 ... and Internet connections
 Using WiFi, a laptop or desktop computer can connect to a network's wireless router
 Bluetooth connects two devices together
 Usually Bluetooth peripherals are powered by batteries that need to be charged or replaced
 WiFi covers a larger distance than Bluetooth
 WiFi requires more than ten times more bandwidth than Bluetooth

[4]

- (c) In order to make reliable information available to all students, schools often put the information on their intranet.

Describe what is meant by an intranet.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (c) Four from:

An intranet is a computer network
 Exists usually within one organisation/school
 Meets the internal needs of an organisation/school
 Can be expanded to become an extranet
 Intranet is private
 Intranets tend to be policed/managed
 Intranet has an extra layer of security
 Data found in an intranet is likely to be more relevant to the students' needs

[4]

- 13 Most school computer networks have a router and switches.

- (a) Explain what is meant by a switch.

1

.....

2

.....

..... [2]

(b) Describe the purpose of a router.

- 1
- 2 [2]

13 (a) Two from:

- Switched hub
- Has many computers connected to it
- Can learn/store addresses of each computer in that part of the network
- Can direct data to specific computers/devices

[2]

(b) Two from:

- Connects network/computers to the internet
- Uses IP addresses
- Transfers data between networks
- Connects LANs/networks together

[2]

15 A school wants to buy some new computers. Some of the computers they buy will be networked and others will be standalone computers which will not be networked.

(a) Describe two advantages of using networked computers.

- 1
- 2 [2]

(b) Describe three disadvantages of using networked computers.

- 1
- 2
- 3 [3]

Eventually the school will use sets of networked Wifi laptops and cabled desktop computers which will also be networked.

(c) Discuss the advantages and disadvantages of using these laptop computers compared with using cabled desktop computers.

[6]

15 (a) Two from:
Fewer printers are needed
Fewer scanners are needed
Can access work from any computer
Data can be shared between computers/data can be accessed by one computer from another more easily
Software can be shared/updated more easily
All computers can access the internet /through one connection [2]

(b) Three from:

- Greater risk of hackers
- Greater risk of viruses
- The significant cost of extra equipment
- When the network is down, cannot use network computers/can still use standalones
- Print queues can be long

(c) Six from:

Advantages

Laptops can be transported from room to room more easily
Safer – won't trip over loose cables
Can use laptops outside the classroom if required
Can be used even if there's a power cut

Disadvantages

Laptops may be more expensive than network PCs.
Display is smaller
Laptops will need recharging periodically
Have to be in range of a network point

One mark available for reasoned conclusion

Must have at least one advantage and disadvantage to gain full marks

[6]

May/June 2014/12

10 A video-conference is being set up.

Complete each sentence below using the most appropriate device from the list.

MICR

OCR

router

web cam

scanner

printer

keyboard

microphone

speaker

(i) Moving images are input using a

.....

(ii) Sound is input using a

.....

(iii) Sound is output using a



.....

(iv) Connection to the Internet is made using a

.....

[4]

10 (i) Webcam

(ii) Microphone

(iii) Speaker

(iv) Router

[4]

19 People can access the Internet using mobile (cell) phones, laptops and their PCs.

Compare and contrast the use of these different devices to access the Internet.

[6]

19 Six from:

- Mobile phones/laptops are portable, PCs are not
- Keyboard on mobile phone is smaller/difficult to type
- Mobile phone display is smaller than laptop or PC
- Mobile phone may not be able to access certain sites
- Signal is likely to be more reliable with PC/laptops as they can be cabled
- Slower to access the internet with mobile phone
- May not always be able to access the internet using a mobile phone or laptop
- More likely to have a mobile phone with you
- Can access the internet from greater number of places with a mobile phone
- Easier to use a mobile phone while on the move
- Some sites do not have full facilities for mobile phones

[6]

2 Identify **two** network devices.

1

2

[2]

2 **Two** from:

Bridge

Hub

Switch

Router

Proxy server

Network card

[2]

10 (a) (i) Describe what is meant by a LAN.

.....

.....

.....

.....[2]

(ii) Describe what is meant by a WAN.

.....

.....

.....

.....[2]

(b) Name an item of hardware that would be used to connect a LAN to a WAN.

..... [1]

- 10 (a) (i) **Two** from:
 LAN is a Local Area Network
 LAN covers a small area/one building
 A school network is a LAN [2]
- (ii) **Two** from:
 WAN is a wide area network
 WAN covers a large geographical area/worldwide
 The Internet is a WAN
 A WAN consists of connected LANs [2]
- (b) Router/modem/cable [1]

16

- (d) To prevent the customers' bank details being read by hackers, the bank uses encryption.

Explain what is meant by encryption.

.....

.....

.....

.....

.....

..... [3]

- (d) **Three** from:
 Causes data to be scrambled/encoded
 Requires an encryption key/software to encrypt
 Requires a decryption key/encryption/key/software to decrypt
 Results in data which is not understandable/readable
 Protects sensitive data...
 ... from being understood if it falls in to the wrong hands [3]

- 9 A number of schools have Local Area Networks (LANs) in their buildings. The LANs are used to share software and for students to save their work.

(a) Describe **two** ways that students can prevent the work they save from being edited by others.

- 1


 2

[2]

The schools want to connect their LANs to the Internet.

(b) Name the device they will need.

..... [1]

As a result of all the schools having access to the Internet, students will be able to communicate with students at other schools in a number of ways.

(c) Give **two** examples of these ways.

- 1

 2

[2]

The headteachers of the schools will want to send confidential data to each other.

(d) Give **one** advantage and **one** disadvantage of encryption.

Advantage

Disadvantage

[2]

- 9 (a) Two from:
 Make their file/work read only
 Password encrypt their file/work
 Grant people access to their file/work without permission to edit
 Make their file/work hidden [2]
- (b) Modem/router [1]
- (c) Two from:
 Email
 Instant messaging
 VOIP
 Video conferencing [2]
- (d) Advantage
 Unauthorised users won't be able to understand data/only person/computer with key can understand data
 Disadvantage
 The data can still be deleted/if the person encrypting the data accidentally deletes the encryption key the data will be unreadable [2]

12 A large multi-national company uses video conferencing instead of face to face meetings between branches in different countries.

- (a) Apart from a keyboard, mouse and display devices, identify **two** other devices which **must** be present in such a system for a video conference to take place.

1

2 [2]

- (b) Describe **four** disadvantages of video conferencing.

1

.....

2

.....

3

.....

4

..... [4]

12 (a) Two from:

Router
Web cam
Microphone
Network card
Speakers

[2]

(b) Four from:

Takes time/costs money to learn new technology/to be trained
Difficult to have international meetings because of time differences
Initial cost of hardware (and software)
Equipment can break down/power cuts can stop conference
Poor strength of signal/time lag/lip sync can be a problem/connection can be lost
Loss of personal/social contact

[4]

October/November

14 Describe the differences between an intranet and the Internet.

.....

.....

.....

.....

.....

.....

.....

..... [4]

14 Four from

Internet is network of networks/intranet doesn't have to be a network of networks
Internet is global
Intranet is within one organisation
Intranet is private/internet is public
Intranets tend to be policed/managed
Intranet has an extra layer of security
Data found in an intranet is likely to be more reliable/relevant than that found on the Internet
Internet has more information than an intranet

[4]



- 16 Describe **three** ways in which personal data held on computers can be misused. For each way explain a different method for helping to prevent it.

Way 1

.....

Method 1

.....

Way 2

.....

Method 2

.....

Way 3

.....

Method 3

.....

[6]

- 16 **Three** matched pairs (with a different method for each one) from:

Data could be amended

Use a username and password so that only the person who knows these can access the data

Use biometrics so that only that person who has those characteristics can access the data

Use a firewall which prevents unknown computers accessing a network

Data could be deleted

Use a username and password so that only the person who knows these can access the data

Use biometrics so that only that person who has those characteristics can access the data

Use a firewall which prevents unknown computers accessing a network

Data could be read and passed on

Encryption so that data is unreadable to unauthorised users

[6]

9 Tick True or False next to each statement.

	True	False
Using a password always prevents unauthorised access to data.		
A strong password is one that is difficult for a hacker to guess.		
Giving your password to a friend is a good idea in case you forget it.		
If you forget your user ID you can still gain access to data using your password.		

[4]

9

	True	False
Using a password always prevents unauthorised access to data		✓
A strong password is one that is difficult for a hacker to guess	✓	
Giving your password to a friend is a good idea in case you forget it.		✓
If you forget your user id you can still gain access to data using your password.		✓

[4]

17 The head teacher of a school is concerned that students may be hacking into the exam scores database to change their marks.

Other than passwords, describe ICT methods that could be used to prevent this from happening.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

17 Four from:

Biometric methods – unique so only authorized users will have access
Encryption makes it difficult for unauthorised users to read data
Firewall – makes it difficult for unauthorised computers to access the system
Disconnect records computer from network – limit access physically
Access levels – only users with appropriate permissions can access data

[4]

May/June 2013/12

4 Tick four devices which are used to connect networks together.

	✓
Hub	
Printer	
Bridge	
Router	
Scanner	
Passwords	
Monitor	
Switch	

[4]

4

	✓
Hub	✓
Printer	
Bridge	✓
Router	✓
Scanner	
Passwords	
Monitor	
Switch	✓

[4]

- 16 A small business wishes to set up a Local Area Network (LAN). The manager is not sure whether to use cables to connect the computers or use wireless technology.

Give **three** disadvantages of using wireless technology.

- 1
- 2
- 3 [3]

- 16 Three from:

Limited area of network
 Strength of signal is weaker
Easier to hack into/less secure
 Physical obstacles can interfere with signal/can cause disconnection [3]

- 17 Describe **four** differences between an intranet and the Internet.

- 1
- 2
- 3
- 4 [4]

17 Four from:

Internet is network of networks/intranet doesn't have to be a network of networks
Internet is global
Intranet is within one organisation
Intranet is private/Internet is public
Intranets tend to be policed/managed
Intranet has an extra layer of security
Data found in an intranet is likely to be more reliable/relevant than that found on the Internet
Internet has more information than an intranet

[4]

May/June 2013/13

14 Mobile phones (cellphones) are used for accessing the Internet.

Give **two** advantages and **two** disadvantages of using a mobile phone rather than using a desktop computer to access the Internet.

Advantage 1

Advantage 2



Disadvantage 1

Disadvantage 2

[4]

14 Two advantages from:

Easy to carry/are portable
Usually have mobile phone in your possession
Can access Internet almost anywhere
Can access Internet on the move

[2]

Two disadvantages from:

Easily lost
May have poorer signal
Display is smaller/keyboard is smaller
Content is more limited
Can be slower to access Internet
Batteries might run out
No mouse so can be more difficult to navigate

[2]

16 School networks often contain a router and a proxy server.

(a) Describe the role of a proxy server in such a network.

.....

.....

.....

.....

.....

..... [3]

(b) Explain the purpose of a router.

.....

.....

.....

.....

.....

..... [3]

16 (a) **Three from:**

- Can act as a web server
- Can act as a buffer (between Internet and LAN)
- Server passes on requests to the Internet
- Passes the requested web pages to individual computers
- Can cache/store the webpages
- Subsequent requests for that/those web page(s) are responded to more quickly
- Can be used to monitor Internet usage
- Can block certain sites

[3]

(b) **Three from:**

- Connects a LAN to a WAN
- Connects a LAN to the Internet
- Forward data packets to the individual computers on a network
- Hold the addresses of each computer on the network

[3]

6 As well as PCs, mobile phones (cellphones) can be used to access the Internet.

(a) Give **two** advantages of using a PC rather than a mobile phone for this purpose.

- 1
- 2
- [2]

(b) Give **two** disadvantages of using a PC rather than a mobile phone for this purpose.

- 1
- 2
- [2]

6 (a) Two from:

Less likely to lose it
 May have stronger signal
 Display is larger/keyboard is larger
 Content is not as limited
 Can be faster to access internet
 No problem with batteries running out
 Has a mouse so is easier to navigate/use [2]

(b) Two from:

Don't always have access to PC/there may be a power cut
 Difficult to carry/not very portable/mobile phone is portable/Cannot access internet except in the house
 You have to buy extra hardware/router [2]

7 Tick **Internet** or **intranet**, next to each statement, as appropriate.

	Internet	intranet
is a network of computer networks		
exists usually within one organisation		
anybody can access it		
can be expanded to become an extranet		

[4]

7

	Internet	Intranet	
is a network of computer networks	✓		[1]
exists usually within one organisation		✓	[1]
anybody can access it	✓		[1]
can be expanded to become an extranet		✓	[1]

5 Name and describe **three** ICT methods which could be used to help prevent **illegal** access to computer networks.

1

.....

.....

.....

.....

2

.....

.....

.....

.....

3

.....

.....

.....

.....

[6]

5 Three matched pairs from:

User ID and one of password/PIN/Memorable data

Password/PIN/Memorable data has to be entered before access is gained/Password/PIN can be changed frequently to avoid hackers guessing them/Unsuccessful logins can throw you out of the system

Magnetic stripe/smart card/electronic key/bar code system/ID card

Has to be read before access is gained/Prevents people without cards accessing system

Biometric data

Fingerprint/retina/iris/face/voice recognition used as input/Biometric data is difficult to replicate/is used because it is unique/compared with those stored on the system

Firewall

Only allows known computers to access network/prevents unknown computers accessing network [6]

9 Describe, including the uses of the hardware and software, what is meant by video conferencing.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[5]

9 Five from:

Video conferencing is a method of visual and audio communication between people at separate locations

Webcams/small video cameras are used to input faces of participants

Large monitor/projector to output faces of participants

Microphones are used to input voices of participants

Speakers are used to output voices of participants

Router/broadband modem used to connect to internet

Video conferencing software needed to link participants

Employees are connected on-line/may need to access the Internet/WAN to work

Images of the employees appear on the screen in real time

Software is needed in each computer to operate the conference

Employees can be heard by all the other employees in the conference

You can take part wherever you are

[5]

7 Multi-national companies often use video conferencing instead of face-to-face meetings.

(a) Tick **three** advantages of video conferencing.

	✓
Companies do not have to buy any equipment	
Companies can arrange the meeting for any time of day	
It is safer as employees do not have to travel	
Companies do not have to pay hotel expenses	
Employees don't have to carry so many documents with them	
Employees are paid less	

[3]

(b) Tick **three** disadvantages of video conferencing.

	✓
Video conferences can only last a short time	
Employees are unable to see who they are talking to	
It is difficult to call international meetings because of time differences	
There is sometimes a time lag between video and sound	
It is difficult to share documents	
Legal documents may need to be signed	

[3]

7 (a)

Companies do not have to buy any equipment	
Companies can arrange the meeting for any time of day	
It is safer as employees do not have to travel	✓
Companies do not have to pay hotel expenses	✓
Employees don't have to carry so many documents with them	✓
Employees are paid less	

[1]

[1]

[1]

(b)

Video conferences can only last a short time	
Employees are unable to see who they are talking to	
It is difficult to call international meetings because of time differences	✓
There is sometimes a time lag between video and sound	✓
It is difficult to share documents	
Legal documents may need to be signed	✓

[1]

[1]

[1]

May/June 2012/11

5 Name three network devices and for each one describe what it does.

Name

Description

.....

Name

Description

.....

Name

Description

..... [6]

5 Three pairs from:**Hub**

Broadcasts data packets to computers in a LAN

Router

Connects LANs to a WAN/Internet

Switch

Directs data packets to specific computers

NIC

Enables computers to be connected to a network

Bridge

Connects networks/LANs together/Directs data packets to specific networks

Proxy server

Stores web pages for faster re-use by computers/can act as a firewall

Modem

Modulates data and demodulates phone signals

[6]

6

	LAN	WLAN
Uses wireless technology to transmit data		✓
Faster transmission of data	✓	
Greater security	✓	
Is cheaper as there is less cabling		✓

[4]

6

	LAN	WLAN
Uses wireless technology to transmit data		✓
Faster transmission of data	✓	
Greater security	✓	
Is cheaper as there is less cabling		✓

[4]

- 14 A bank is looking into a variety of security measures to protect its customers' data. Describe **three** types of authentication technique they could use and for each give a benefit and a drawback.

Technique 1

Benefit

Drawback

Technique 2

Benefit

Drawback

Technique 3

Benefit

Drawback

[9]

14 Three matched triples from:

User ID and password

Benefit – each user ID could be unique/only user will know the password/data can only be accessed by person who knows the password/Password can be changed frequently to avoid hackers guessing them/Unsuccessful logins can throw you out of the system

Drawback – user might forget password/keylogging software can be used to intercept it

Biometrics

Benefit – each user has unique biometrics

Drawback – equipment is expensive to buy/may be difficult or expensive to get equipment to user

Magnetic/chip card with PIN

Benefit – Hacker needs to have the card and know the PIN

Drawback – can lose the card/can forget PIN

TAN

Benefit – Always changing so a hacker would not be able to use it even if they intercepted it when user typed it in.

Drawback – need to have card and remember PIN and use it within a short period of time. [9]

14 Mobile telephones (cellphones) are used for accessing the internet.

Give **two** advantages and **two** disadvantages of using a mobile telephone rather than using a laptop computer to access the internet.

Advantage 1

.....

Advantage 2

.....

Disadvantage 1

.....

Disadvantage 2

..... [4]

- 14 Two advantages from:
 More likely to have it with you
 Can access internet from greater number of places
 Easier to use while on the move

Two disadvantages from:
 Display is more difficult to see/display screen is smaller
 Keyboard is smaller so more difficult to type
 Some sites do not have full facilities for mobile phone

[4]

- 16 Describe what a virus is and what effect it may have on the data on a hard disc.

.....

.....

.....

.....

.....

.....

.....

[3]

- 16 Three from:
 A piece of program code
 Which replicates itself
 Fills up hard disc making it unusable
 Deletes data/changes data (from hard disc)
 Makes software/operating system unusable

[3]

- 8 Tick **True** or **False** next to each statement describing encryption.

	True	False
Encryption prevents hackers from understanding the data.		
Encryption destroys viruses.		
Encryption prevents unauthorised access to computer systems.		
Encryption is the scrambling of data.		

[4]

8

	True	False
Encryption prevents hackers from understanding the data	✓	
Encryption destroys viruses		✓
Encryption prevents unauthorised access to computer systems		✓
Encryption is the scrambling of data	✓	

[4]

- 11 A manager of a new company has just purchased some computers. She wishes to connect these computers together to form a LAN.
Name and describe **three** network devices she might need to have in order to achieve this.

Device 1

Description

.....

Device 2

Description

.....

Device 3

Description

.....

[6]

- 11 Three matched pairs from:

Hub

Broadcasts data packets to computers in a LAN

Switch

Directs data to specific computers

Bridge

Connects two LANs together to form a larger LAN/Directs data packets to specific networks

NIC

Enables computer to be connected to a network

[6]

- 14 Schools are concerned with the possibility that students are using the schools' internet connections to access undesirable sites.

Name and describe **two** security methods the schools could use to prevent this from happening.

Method 1

Description

Method 2

Description

[4]

14 Two matched pairs

Proxy server

The proxy server can restrict the web sites students can access/can block access to sites with objectionable material

Firewall

Limits the computers (using IP address) that can access the system/that can be accessed from within the school

Applying parental guidance settings.

Can block undesirable topics

[4]

6 Ali wants to back up his data.

(a) Give **two** reasons why making backups is necessary.

- 1
-
- 2
- [2]

(b) Ali thinks that making backups will prevent his data from getting viruses. Describe what a virus is and explain why Ali is wrong.

-
-
-
-
- [3]

6 (a) **Two** from:

Data could be corrupted on original medium
 Data could be accidentally amended/overwritten on original medium
 Data could be accidentally deleted from original medium
 Data could be maliciously deleted from original medium

[2]

(b) **Two** from:

Virus is a piece of programming code
 Which replicates itself...
 Can fill up storage medium making it unusable
 Deletes data/change data
 Makes software/operating system unusable

[2 max]

If the original medium has a virus the backup will have it as well/backup will have no effect on the hard disc so it is still susceptible to viruses

[1]

10 Describe the benefits and drawbacks to companies of using video conferencing.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[5]

10 Five from:

Benefits:

- Can call meeting at short notice
- No need to pay travelling expenses
- Can work from home
- Employees will save time travelling
- Do not have to pay hotel expenses
- Do not have to pay for conference room facilities
- Safer as participants do not have to travel
- Don't have to transport/carry equipment/lots of documents/documents don't get lost in transit

Four max.

Drawbacks:

- Takes time/costs money to learn new technology/to be trained
- Difficult to have international meetings because of time differences
- Initial cost of hardware (and software)
- Equipment can break down/power cuts can stop conference
- Poor strength of signal/time lag/lip sync can be a problem/ connection can be lost
- Loss of personal/social contact
- Legal documents cannot be signed

Four max.

[5]

6 Name three items of hardware, other than a PC, needed to conduct a video conference.

- 1
- 2
- 3 [3]

6 Three from:

Projector/large screen
Webcam/video camera
Speakers/headphones
Microphone
Router/modem

[3]

9 Aftab owns a company which has a small number of computers. He wants to connect the computers into a network. He does not wish to connect the computers to the internet but he does want the workers to be able to communicate within the company.

(a) Name a network device used to connect the computers together.

..... [1]

(b) Name an item of software used to send messages between the computers.

..... [1]

(c) What is the name given to this type of network?

..... [1]

(d) Give three reasons why Aftab may not want his workers to access the internet.

- 1
- 2
- 3
- [3]

- 9 (a) Hub/switch/bridge [1]
- (b) Email software/IM software [1]
- (c) LAN/intranet [1]
- (d) Three from:
- They may waste time playing games/going on social network sites/updating their blog/ online shopping/banking/surfing
 - They could access undesirable sites
 - They could download viruses/malware
 - Could make the company susceptible to hackers
- [3]

October/November

2 Ring two network devices.

CD RW

hard disc

hub

PIN

plotter

switch

[2]

- | | | | | |
|---|-------|-----------|--------|-----|
| 2 | CD RW | hard disc | hub | [2] |
| | PIN | plotter | switch | |

11 Many banks offer internet banking. To access this online service customers have to log on to the system.

(a) Explain why authentication techniques are necessary.

.....

.....

.....

.....

.....

.....

[3]

(b) Customer data is encrypted. Explain what encryption is and why it is used.

.....

.....

.....

.....

.....

..... [3]

11 (a) Three from:

Authentication techniques such as user names and passwords identify the user to the system
Without authentication anybody would be able to access data
Hackers would be able to amend/delete data without being prevented
Would have to rely on other methods such as firewalls to prevent unauthorised access. [3]

(b) Three from:

Causes data to be scrambled/encoded
Requires an encryption key/software to encrypt
Requires a decryption key/encryption software to decrypt
Results in data which is not understandable/readable
Protects sensitive data...
... from being understood if it falls in to the wrong hands [3]

- 16 Some mobile telephones (cell phones) can be used to make video calls.
Describe **four** advantages of making video calls rather than sending messages by email.

1
.....
2
.....
3
.....
4
..... [4]

16 Four from:

You get immediate feedback
You know you have the right person/don't have to worry that you've got the wrong email address
You can ask questions immediately based on feedback/carry out an interactive conversation
Less impersonal/less risk of upsetting recipient
Can make yourself understood more easily
You can see the other person's reaction [4]